

10 — Liquid Track: Concept and Physics

v1.1 — CaCl₂ brine dehumidification with mixed-mode ventilation

Function: The low-energy, solar-grade comfort layer: hold a ~100 m³ occupied space at ~55% RH for four adults at DP-A on 25–80 W of electricity and ~9–11 kWh/day of 60–93 °C heat, with berth-scale evaporative spot cooling and a moisture battery. Shared basis, design point, and doctrines → doc 00.

1. The problem at the design point

DP-A (32 °C / 80%, ω 24.2 g/kg, dew point 28.1 °C, raw-water sink ~29 °C). Comfort target **29 °C / 55% RH** → ω = **13.8 g/kg**. Occupant latent load ≈ 70 g/h·person average (50 sleeping, 90 active). Conventional answers fail the constraints:

- **Vapor-compression A/C:** 1–3 kW electrical → generator/grid dependence, a refrigerant circuit, and (marine) a salt-air-hostile machine.
- **Sub-dew-point condensation** is handicapped: the raw-water sink is *warmer than the dew point*, so nothing passively cold exists to condense on (doc 00 §3).
- **Solid desiccant wheels** regenerate with 300–600 W electric heaters.
- **Small heat-driven chillers** are commercially extinct below ~11 kW.

2. Operating principle

Vapor-pressure depression does the drying — no cold surface required.

1. Concentrated CaCl₂ brine at **40 wt%** has water activity **aw ≈ 0.45**: air in contact equilibrates near 45% RH *at the brine's temperature*. With the sink at 29 °C the brine runs 29–30 °C → **outlet floor ~11.9 g/kg** (design basis); the outlet is a **temperature-dependent band, not a point** (floor matrix, doc 12).
2. **Mixed-mode air path (occupied baseline, X6):** fresh air at the CO₂-governed minimum (48 m³/h for 4, through the $\epsilon \geq 0.8$ ERV) plus **recirculated cabin air (~75 m³/h)** together pass through the brine contactor — ~123 m³/h total. The recirculation decouples removal capacity from the ventilation ration; the fresh minimum and interlock are fixed by doc 00 §5. Positive pressure is held; recirc-only operation is prohibited while occupied. Once-through operation is retained for unattended/mothball and low-occupancy modes (2 adults with cooled brine: ~54–58% RH — stands on the base system).
3. Absorbed water dilutes the brine toward **35 wt%**; each kg of 40 wt% concentrate absorbs **0.143 kg water** (0.40/0.35 – 1).
4. A peristaltic pump trades dilute brine (0.04–0.2 L/min) to the **regenerator**: heat at 60–93 °C raises the brine's vapor pressure far above the sink-cooled condenser's (aw 0.55 at 65 °C → Pv ≈ 13.8 kPa vs ~4–5 kPa) and the water leaves — into a **sealed still** as distillate (primary), or a swept airstream (degraded mode, now condensed per X8 rule 3).
5. Regenerated concentrate returns to reserve. **Concentration is storage.**

The moisture battery

At DP-A occupied duty (0.88 kg/h peak, doc 12 §1) a full night of 4-adult operation with zero heat input costs **~35–40 kg of 40 wt% concentrate** (12 h ÷ 0.143, duty-averaged) — inside the 25–55 kg reserve spec, noted for tank sizing. Regenerate by day (solar) or from engine/waste heat; absorb any time. A storm that defeats the absorber is an hours-to-a-day event — shorter than the reserve. The CO₂ battery (doc 00 §5) mirrors this pattern on the air-quality side.

The closed water loop (Phase 2)

The still's distillate feeds a **Maisotsenko-cycle (M-cycle) dew-point indirect evaporative cooler**. M-cycle cooling is useless on raw DP-A air (dew point 28.1 °C is its floor) but potent on dried air:

Feed air	Dew point	Supply (ϵ_{dp} 0.65–0.80)
11.9 g/kg (base CaCl ₂ , cooled brine)	16.7 °C	~21–23 °C
9.0 g/kg (hot regen) / 7.5–8 (LiCl blend)	12.5 / ~10 °C	~18–21 / ~15–18 °C

Full chain: **sun → hot brine → dry air + distilled water → cool dry air.**

Berth-scale cascade (the adopted Phase-2 topology, X5): the M-cycle rides the ventilation stream — all 48 m³/h delivered to the cabin as product; the working air (⅓) is drawn *from the cabin* and exhausted saturated. At DP-A: supply ~22.6 °C, ~102 W sensible at the outlet, full ventilation preserved, water draw ~4.1 L/day (≤ distillate). The classic dry-channel bleed is rejected: it silently exhausts a third of the ventilation (CO₂ → ~2,700 ppm) for less delivered cooling (~86 W). **Whole-cabin M-cycle AC is deferred** — X1: it never composes with once-through (5.5–11 kg/h absorber duty), and in recirculation topology it converges to solid-track-class duty (~10.6 kg/h, ~11 kW heat; doc 00 §6).

3. Governing quantities (the model in nine lines, DP-A)

Quantity	Relation	Design value
Removal rate, mixed-mode 4 adults	$\Sigma \text{ flow} \times (\omega_{\text{in}} - \omega_{\text{floor}})$	0.88 kg/h peak (fresh 58 kg/h from 24.2 pre-dried 17.4 by ERV, + recirc from 13.8, to floor 11.9)
Water per kg concentrate	$c_{\text{hi}}/c_{\text{lo}} - 1$	0.143 kg (40→35 wt%)
Transfer brine flow	$\text{removal} \div 0.143 \div p$	0.07 avg / 0.35 peak L/min
Regen latent duty	$\text{removal} \times 2.44 \text{ MJ/kg}$	0.60 kW at 0.88 kg/h
Regen heat at COP 0.55–0.75	$\text{latent} \div \text{COP}$	~0.92 kW peak; 17–20 kWh/day bare, ~9–11 with ERV + DCV
Absorption heat into brine	$\text{removal} \times \sim 2.7 \text{ MJ/kg}$	0.66 kW — must be rejected (raw-water coil)
Cooling leverage	$\partial w_{\text{eq}}/\partial T$ at aw 0.45	0.5–0.7 g/kg per °C of brine cooling (land sinks: free depth)
CO ₂ battery heat (doc 00 §5)	$1.6 \text{ kg/day} \times 1.0\text{--}1.3 \text{ kWh/kg}$	~1.6–2.5 kWh/day
Crystallization liquidus	CaCl ₂ solubility curve	40 wt% $\approx$ 12–13 °C ; 44 wt% $\approx$ 22 °C

Open item (doc 12 §2, erratum 10): the ERV pre-dried figure of 17.4 g/kg in the removal row above corresponds to $\epsilon_{\text{lat}} \approx 0.65$, while doc 12 §1 states 15.9 g/kg at the specified $\epsilon \geq 0.8$. The two are not reconcilable at one effectiveness, and the published ERV'd duty follows the 17.4 value. Left standing and flagged rather than silently corrected; **test E measures the real ϵ_{lat} and settles it.**

Two facts dominate system character:

- **Absorber cooling is required, not optional.** 0.66 kW enters the sump; the air stream removes only ~35 W/K **at the ~123 m³/h mixed-mode flow** (~16 W/K on the 48 m³/h fresh stream alone — always state which flow a W/K figure is on), so an uncooled contactor self-limits by warming +5–8 K and doing less work (floor 12–14+ g/kg), ~300 L/h of raw water through a titanium coil (5–15 W pump) holds the design floor.
- **COP stops mattering when heat is generous.** With engine-jacket, waste, or oversized solar heat, the binding constraints become materials temperature ceilings, contactor area, and heat rejection — not energy. At 85–93 °C regeneration (within CPVC's rating) plain CaCl₂ reaches 43–44 wt% (aw 0.33–0.35) → floor **7.5–9 g/kg** without LiCl.

4. Why these choices (decision record)

Decision	Over	Because
Liquid desiccant	vapor-compression	No compressor; chloride-immune by construction; heat-driven (solar/diesel/engine/waste all serve); storage free as concentration. The wins are resilience and heat flexibility, not watts
Liquid	solid beds	Pumpable, continuous, motion-tolerant; deliquescence becomes the operating principle instead of the failure mode
Film (trickle) media primary	bubble columns	~90% electrical cut (5–100 Pa vs 1.4–2.8 kPa air-side); columns retained as annex for sustained heel; storm fallback is flooded-media mode , not gimballs
CaCl ₂	LiCl / LiBr / MgCl ₂ / glycols	Adequate depth at 1/20–1/50 LiCl's cost → cheap oversized storage. LiCl remains a comfort-spike upgrade; glycols excluded (vapor carryover into breathed air)
Mixed-mode (fresh minimum + recirc)	once-through-only	Once-through fails 4-adult comfort at DP-A (59–66% RH) and couples removal to the ventilation ration; mixed-mode restores 55% RH on base brine at 0.88 kg/h with CO ₂ unchanged at the interlocked floor. Once-through retained for unattended and low-occupancy duty, where per-pass depth on the wettest air still wins
ERV latent recovery on the fresh stream (X7)	bare fresh intake	$(1-\epsilon) \times 10.4 \text{ g/kg}$ per kg of fresh air; at $\epsilon \approx 0.8$ the DP-A fresh-air penalty drops ~9–10 kWh/day and generous ventilation becomes cheap. Requires one ducted exhaust path (semi-balanced flow)
Buffer-tank thermal bus	direct coupling	Merges solar + hydronic + engine/waste heat into one 65–90 °C supply; kills short-cycling; provides DHW
Sealed-still regeneration	air-swept	Deletes regen blower, demister, salty-exhaust routing; captures distillate; most motion-tolerant regenerator possible. Air-swept retained as degraded mode, now condensed (X8)

5. Performance envelope (DP-A)

Scenario	Outcome
4 adults, once-through, base brine	59–66% RH — fails ; mode retired for full occupancy
4 adults, mixed-mode, cooled base brine (baseline)	~55% RH at 29 °C · 0.88 kg/h · 9–11 kWh/day with ERV
2 adults, once-through, cooled brine	~54–58% RH — stands
Unattended, 0.1 ACH, 1 contactor	mold-safe to ~160–215 m³ (60–65% RH basis) / ~90 m ³ holding 13 g/kg, on ~30–50 W + trickle of heat
Berth cascade M-cycle	~22.6 °C supply, ~102 W/outlet, full ventilation, ~4 L/day water

